

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Review/Challenge Questions

1. A series has a sum if what sequence has a limit?

2. Consider an infinite series $\sum_{n=1}^{\infty} a_n$.

a) Suppose that $\sum_{n=1}^{\infty} a_n = 3$. What can you say about the sequence $\{a_n\}$?

b) Suppose that $\lim_{n \rightarrow \infty} a_n = \frac{1}{3}$, what can you say about $\sum_{n=1}^{\infty} a_n$?

3. Find the sum of the following series or explain why the sum does not exist.

a) $1 + \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots + \frac{1}{3^n} + \dots$

b) $1+2+4+8+\dots+2^n+\dots$

c) $\sum_{i=4}^{\infty} \frac{3}{i}$

4. Determine the sum of the following series.

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{9} + \dots + \left(\frac{1}{2^n} + \frac{1}{3^n} \right) + \dots$$

5. Can you use the Ratio Test to determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^4}$. Why or why not?
6. ($n!$ versus 2^n) Show that $\sum_{n=1}^{\infty} \frac{2^n}{n!}$ converges. What does this tell you about $\lim_{n \rightarrow \infty} \frac{2^n}{n!}$? Which sequence goes faster to zero, $\left\{ \frac{1}{2^n} \right\}$ or $\left\{ \frac{1}{n!} \right\}$?
7. Determine whether the following series are (absolutely or conditionally) convergent, or divergent.
- a) $\sum_{n=1}^{\infty} \frac{n^2 + 1}{n^2 + 4}$

$$\text{b) } \sum_{n=2}^{\infty} \frac{1}{n^2-1}$$

$$\text{c) } \sum_{n=1}^{\infty} \frac{n!}{n^2}$$

$$\text{d) } \sum_{n=1}^{\infty} \frac{(n!)^n}{(n^4)^n}$$

e) $\sum_{n=1}^{\infty} (-1)^n \frac{\ln(n)}{n}$

f) $\sum_{n=1}^{\infty} \frac{3^n}{3^n - 2^n}$

8. For what values of x does the series absolutely converge?

a) $\sum_{n=0}^{\infty} (x-2)^n$

b) $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$

9. Show that the following series diverges except for exactly one value of c and compute the sum of the series for that value of c .

$$\sum_{n=2}^{\infty} \frac{cn+1}{n^2-n}$$